# ELEC3104: Digital Signal Processing

## Topic 4: Digital Resonator Fundamentals

- Second-Order Digital Resonator (Ref pp. 343-348)
- Stability of a Second - Order Filter (pp. 205-209)
- Digital Oscillators (Ref pp. 355-357)

Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

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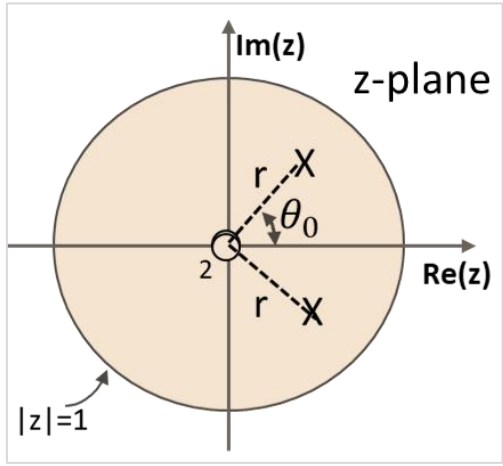
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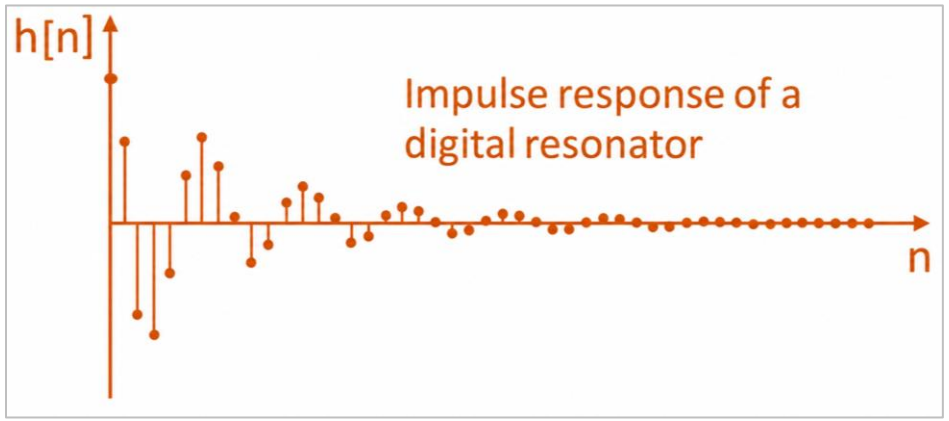
# Digital Resonator Fundamentals

Introduction

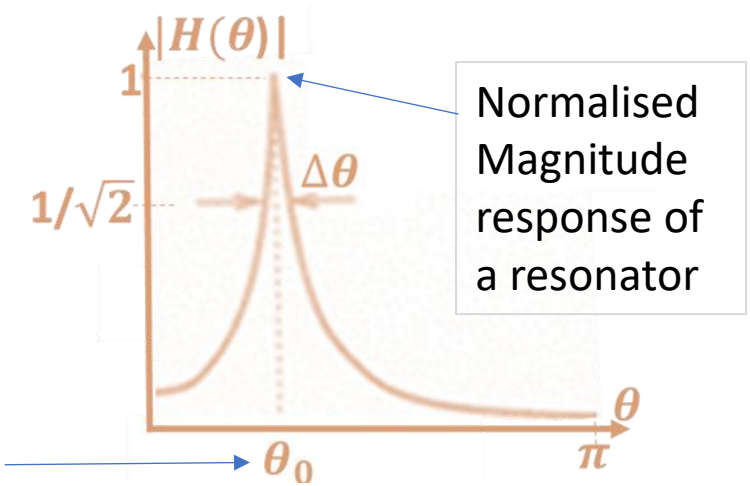
- A digital resonator is typically a two-pole IIR filter with a complex-conjugate pair of poles inside the unit circle in the z-plane and two zeros at $z = 0$.
- The pole angle $\theta_0$ approximately determines the resonant frequency, while the pole radius $r$ measured from the origin, controls the bandwidth $\Delta\theta$.
- As the poles approach the unit circle, the bandwidth decreases and the resonance peak becomes sharper.
- A digital resonator acts as a narrowband bandpass filter.




The frequency of the magnitude peak is approximately the pole angle; they are not exactly equal in general




Normalised Magnitude response of a resonator

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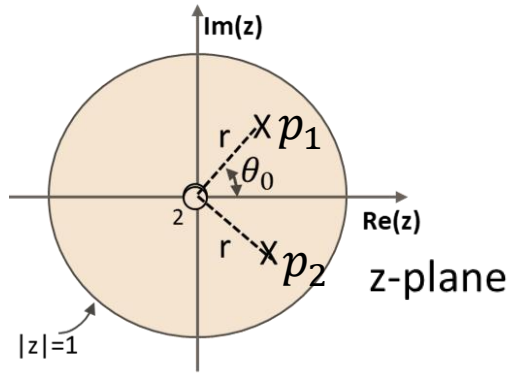
# A second-order digital resonator

## A second-order digital resonator

- One way to realise a second-order resonator with a peak approximately at $\theta_0$ with a complex-conjugate pair of poles at angles $\pm\theta_0$ ($p_1 = re^{j\theta_0}$ and $p_2 = re^{-j\theta_0}$) and a pair of zeros at $z = 0$. (see diagram on the right)
- To ensure stability , we choose the pole radius r to be less than (but close to) unity.
- The transfer function of such a digital resonator with zeros at the origin is:

$$H(z) = \frac{z^2}{(z-p_1)(z-p_2)} = \frac{z^2}{(z-re^{j\theta_0})(z-re^{-j\theta_0})}$$

$$H(z) = \frac{z^2}{z^2 - r(e^{j\theta_0}+e^{-j\theta_0})z + r^2} = \frac{z^2}{z^2 - 2r\cos\theta_0 z + r^2} \quad \ldots\ldots\ldots (1)$$

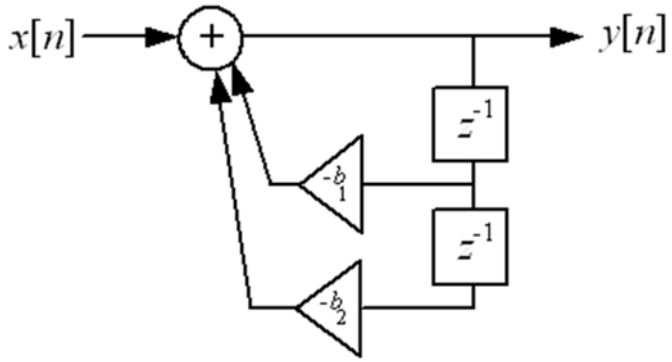


z-plane

$$p_1 = re^{j\theta_0} = r\cos\theta_0 + jr\sin\theta_0$$
$$p_2 = re^{-j\theta_0} = r\cos\theta_0 - jr\sin\theta_0$$

From the second- order realisation (see diagram on the right), we have

$$H(z) = \frac{1}{1 + b_1 z^{-1} + b_2 z^{-2}} = \frac{z^2}{z^2 + b_1 z + b_2} \ldots. (2)$$



A second-order structure

Equations (1) and (2) are equivalent. By equating the coefficients, we obtain

$$b_1 = -2r\cos\theta_0 \quad \text{and} \quad b_2 = r^2; \quad \therefore \cos\theta_0 = -\frac{b_1}{2\sqrt{b_2}} \qquad \theta_0 = \frac{2\pi f_0}{f_s}$$

# Digital Resonator: Magnitude Response

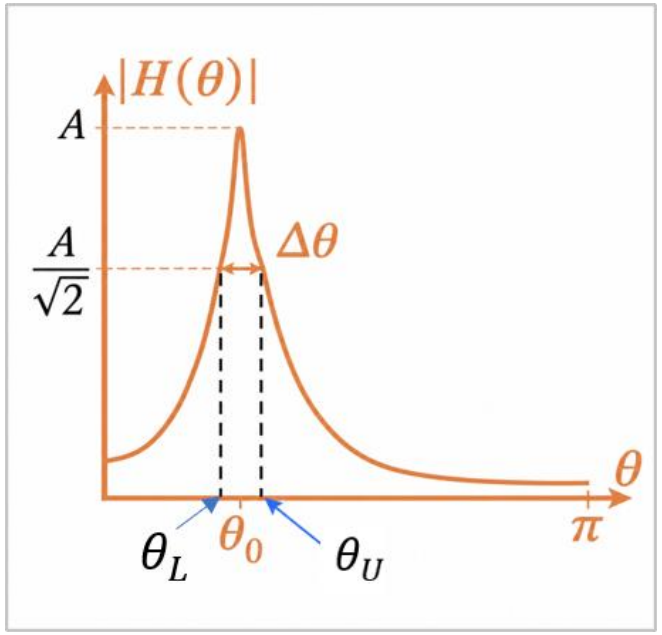
## Magnitude Spectrum

- The digital resonator's magnitude squared function $|H(\theta)|^2$ can be obtained from H(z) as follows:

$$H(z) = \frac{z^2}{z^2 - 2r\cos\theta_0 z + r^2}$$

$$|H(\theta)|^2 = \left|\frac{e^{j2\theta}}{e^{j2\theta} - 2r\cos\theta_0 e^{j\theta} + r^2}\right|^2 = \frac{1}{[1-2r\cos(\theta-\theta_0)+r^2]\,[1-2r\cos(\theta+\theta_0)+r^2]}$$

- For r close to unity , the magnitude peak occurs approximately at $\theta_0$.
- The half-power frequencies $\theta_L$ and $\theta_U$ satisfy $|H(\theta)| = {}^{A}/_{\sqrt{2}}$ where A is the actual peak magnitude.
- The bandwidth is $\Delta\theta = \theta_U - \theta_L$ .




The peak gain can be normalised to unity by dividing the transfer function *H(z)* by A.

# Digital Resonator: Bandwidth Approximation

## Bandwidth Approximation

- We assume that the bandwidth is very small: $\theta = \theta_0 + \delta$ where $\delta$ is a small frequency offset. Therefore $\theta + \theta_0 = 2\theta_0 + \delta \approx 2\theta_0$
- The half-power frequencies lie approximately symmetrically about $\theta_0$:

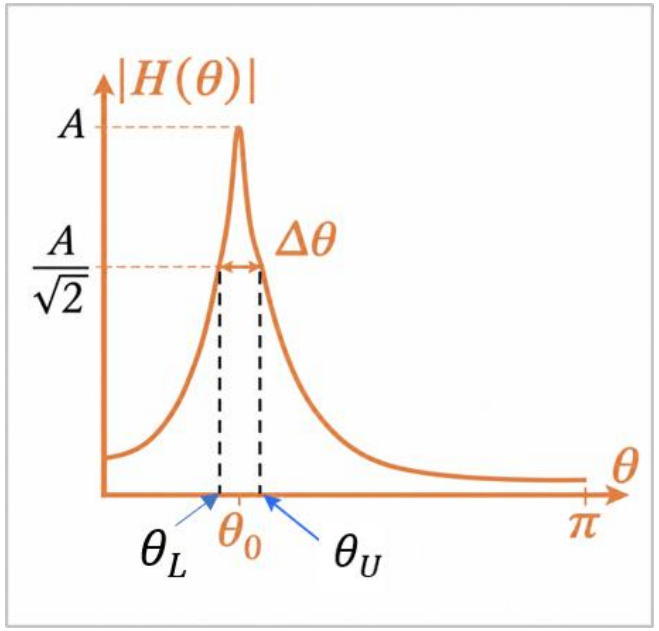
$$\theta_L \approx \theta_0 - \frac{\Delta\theta}{2}, \qquad \theta_U \approx \theta_0 + \frac{\Delta\theta}{2}$$

- Thus, at the half-power frequencies, $\theta - \theta_0 \approx \pm\frac{\Delta\theta}{2}$
- This symmetry is an approximation; the actual peak and half-power frequencies need not be exactly centred on $\theta_0$.
- The magnitude squared response is:

$$|H(\theta)|^2 = \frac{1}{[1-2r\cos(\theta-\theta_0)+r^2]\ [1-2r\cos(\theta+\theta_0)+r^2]}$$

- At $\theta = \theta_0$ (or very close to) the resonant frequency ($\theta_0$)

$$|H(\theta_0)|^2 = \frac{1}{[(1-r)^2]\ [1-2r\cos(2\theta_0)+r^2]}$$



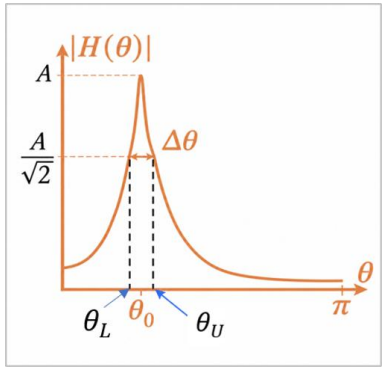
# Digital Resonator: Bandwidth Approximation

## Bandwidth Approximation

- The half-power bandwidth $\Delta\theta$ is found by locating the frequencies at which $|H(\theta)|^2 = 0.5\,|H(\theta_0)|^2$

$$\therefore \frac{1}{[1-2r\cos(\theta-\theta_0)+r^2]\;[1-2r\cos(\theta+\theta_0)+r^2]} = \frac{0.5}{[(1-r)^2]\;[1-2r\cos(2\theta_0)+r^2]} \rightarrow \frac{1}{1-2r\cos(\theta-\theta_0)+r^2} \approx \frac{1}{2(1-r)^2}$$

$$\therefore 1 - 2r\cos(\theta-\theta_0) + r^2 \approx 2$$




- Substituting $\theta - \theta_0 \approx \pm\frac{\Delta\theta}{2}$, $\boxed{1 - 2r\cos\left(\frac{\Delta\theta}{2}\right) + r^2 \approx 2(1-r)^2}$.
- Using Taylors expansion of cosine , we obtain: $\cos\left(\frac{\Delta\theta}{2}\right) \approx 1 - \frac{12}{\left(\frac{\Delta\theta}{2}\right)}^2 = \boxed{1 - \frac{\Delta\theta^2}{8}}$ and substitute this into the half-power equation:
- $1 - 2r\left(1 - \frac{\Delta\theta^2}{8}\right) + r^2 \approx 2 \rightarrow 1 - 2r + r^2 + \frac{r\Delta\theta^2}{4} \approx 2 \rightarrow \therefore \frac{r\Delta\theta^2}{4} \approx (1-r)^{2\cdot} \rightarrow \Delta\theta \approx \frac{2(1-r)}{\sqrt{r}}$
- When $r$ is very close to unity $\sqrt{r} \approx 1,\ giving \quad \Delta\theta \approx 2(1-r) \quad \Longrightarrow \quad \boxed{r \approx 1 - \frac{\Delta\theta}{2}}\ ; r \approx 1 - \pi\frac{\Delta f}{f_s};\ \Delta\theta = 2\pi\frac{\Delta f}{f_s};$
- We can show that the $A \approx \frac{1}{(1-r^2)\sin(\theta_0)}$

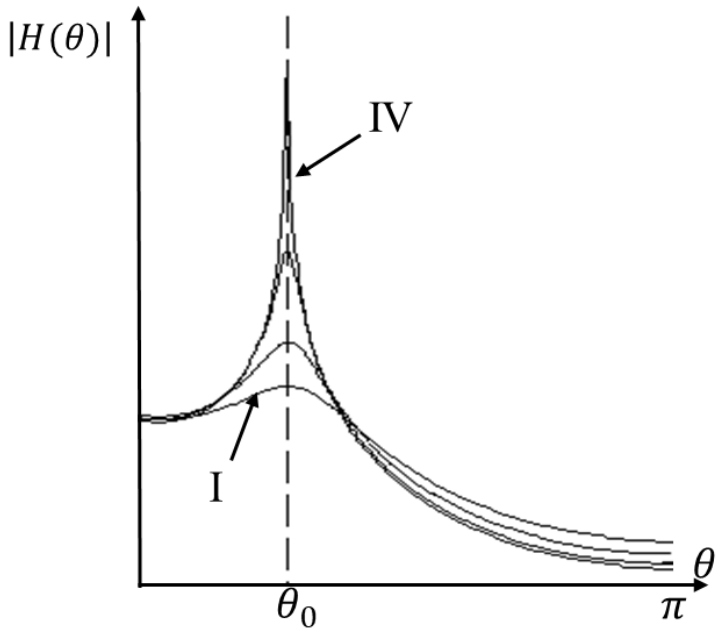
# Digital Resonator Fundamentals

## Magnitude Spectrum

- From the resonator transfer function H(z), we can derive the magnitude response as discussed previously:

$$H(z) = \frac{1}{1+b_1 z^{-1}+b_2 z^{-2}}; \quad b_2 = r^2; \quad b_1 = -2r\,cos(\theta_0); \quad \theta_0 = cos^{-1}\left[\frac{-b_1}{2\sqrt{b_2}}\right]$$

- The magnitude spectrum is shown below for various values of $b_1$ and $b_2$




| | $b_1$ | $b_2$ |
|---|---|---|
| I | -0.94 | 0.5 |
| II | -1.16 | 0.7 |
| III | -1.34 | 0.9 |
| IV | -1.41 | 0.99 |

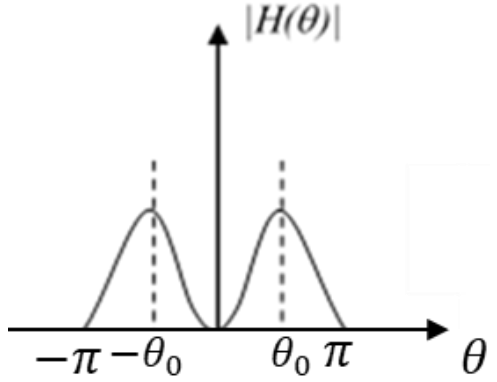
As $b_2 = r^2$ approaches unity, the resonance becomes sharper, the bandwidth decreases and the peak gain increases

**Note:** The pole angle $\theta_0$ and the actual peak frequency $\theta_p$ where the magnitude response reaches its maximum are approximately equal when the poles are close to the unit circle

To make the magnitude response zero at $\theta = 0$ and $\theta = \pi$, place the two zeros at z = 1 and z = -1. This gives the modified transfer function $H_1(z)$ .

$$H_1(z) = \frac{1 - z^{-2}}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

Its magnitude response is shown below:




Magnitude response

# Digital Resonator Design

Example 1

Design a digital resonator with a peak gain of unity ay 50Hz and a 3-dB bandwidth of 6 Hz , assuming the sampling frequency of 300Hz.

- The resonant frequency $\theta_0 = \frac{2\pi(50)}{300} = \frac{\pi}{3}$;
- The 3-dB bandwidth $\Delta\theta = \frac{2\pi(6)}{300} = 0.04\pi$.
- The pole radius $r \approx 1 - \frac{\Delta\theta}{2} = 1 - 0.02\pi \approx 0.9372$.
- The transfer function of the digital resonator H(z) is:

$$H(z) = \frac{k_0 z^2}{z^2 - 2r\cos\theta_0 z + r^2} = \frac{k_0 z^2}{z^2 - 0.9372z + 0.8783}$$

$\theta_0 = \frac{\pi}{3}$ is the pole angle corresponding to the desired resonant frequency. The actual peak is very slightly below 50 Hz.

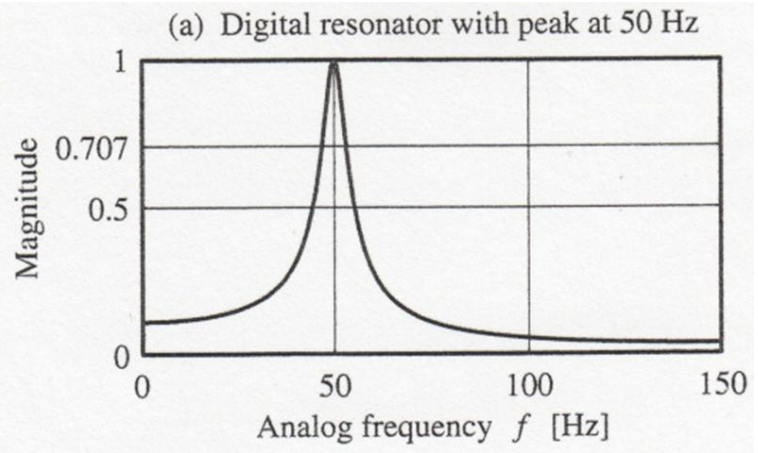
- For a peak gain of unity, we choose $k_0 = \frac{1}{A} = (1 - r^2)\sin(\theta_0) = 0.1054$

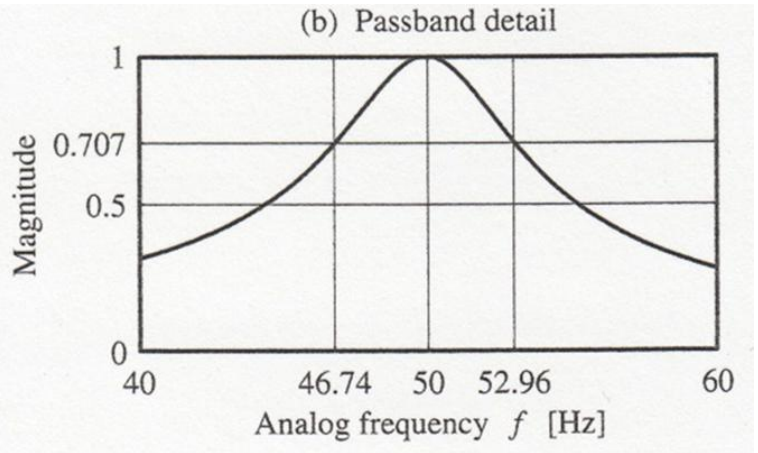
$$\therefore H(z) = \frac{0.1054\, z^2}{z^2 - 0.9372z + 0.8783} = \frac{0.1054}{1 - 0.9372z^{-1} + 0.8783z^{-2}}$$

- The magnitude spectrum and the passband details are shown on the right
- The passband detail shows that the half-power frequencies are approximately 46.74 Hz and 52.96 Hz. The resulting 3-dB bandwidth is

$$\Delta f = 52.96 - 46.74 = 6.22 \text{ Hz}$$

Which is close to the specified 6Hz. The difference between 6.22 Hz and 6 Hz illustrates the bandwidth approximation used in the design.



# Digital Notch filters : Pole-Zero Placement

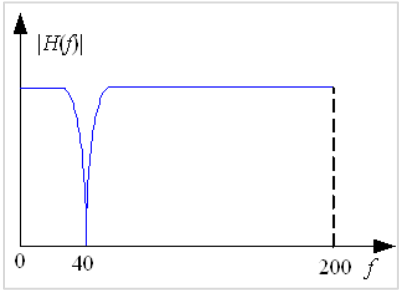
## Notch filters

- When a zero is placed at a given point on the unit circle, the frequency response will be zero at the corresponding point.
- A pole on the other hand produces a peak at the corresponding frequency point.
- Poles that are close to the unit circle give rise to large peaks, whereas zeros close to or on the unit circle produce troughs or minima.
- Thus, by strategically placing poles and zeros on the z-plane, we can obtain simple low pass or other frequency selective filters (**notch filters**).

## Example 2:

Obtain, by the pole-zero placement method, the transfer function of a simple digital notch filter (see figure below) that meets the following specifications:

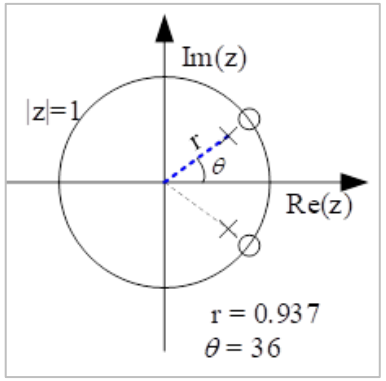
- Notch Frequency: 40Hz
- 3db notch width: 8 Hz
  (approx.: ±4Hz about 40Hz)
- Sampling frequency: 400 Hz




The radius, r of the poles is determined by: $r = 1 - \left(\frac{\Delta f}{f_s}\right)\pi$

- To reject the component at 40 Hz, place a pair of complex zeros at the points on the unit circle corresponding to ±40 Hz. i.e. at an angles: $360^{\bullet} \times \frac{40}{400} = \pm 36^{\bullet} = \pm 0.2\pi$
- To achieve a sharp notch filter and improved magnitude response either side of the notch frequency, a pair of complex conjugate poles are placed at a radius $r < 1$, at the same angle as the zeros, as shown below.
- $r = 1 - \left(\frac{\Delta f}{f_s}\right)\pi = 1 - \left(\frac{8}{400}\right)\pi = 0.937$

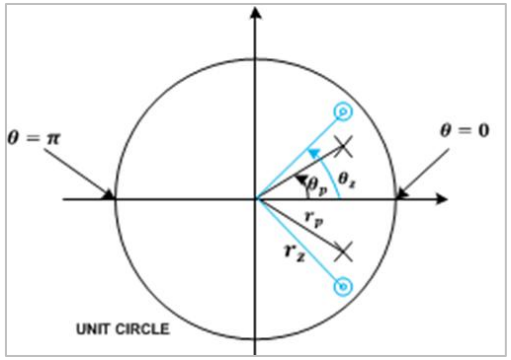
$$H(z) = \frac{K\left(z - e^{-j0.2\pi}\right)\left(z - e^{j0.2\pi}\right)}{\left(z - 0.937e^{-j0.2\pi}\right)\left(z - 0.937e^{j0.2\pi}\right)}$$




$$H(z) = \frac{K\left(1 - 1.6180z^{-1} + z^{-2}\right)}{1 - 1.5161z^{-1} + 0.878z^{-2}}$$

# Digital Filters: Pole zero placement

**Question:** Determine the digital filter coefficients by placing poles and zeros at the locations shown on the right




- ✓ Pole frequency = $\theta_p$; radius = $r_p$; $\theta_p = \frac{2\pi f_p}{f_s}$; $f_s = 16kHz$ (or higher)
- ✓ zero frequency = $\theta_z$; radius = $r_z$ ($r_z > r_p$ and closer to unit circle); $\theta_z = \frac{2\pi f_z}{f_s}$ $[\theta_z > \theta_p]$
- ✓ $H_p(z) = \frac{z^2}{\left(z - r_p e^{j\theta_p}\right)\left(z - r_p e^{-j\theta_p}\right)} = \frac{z^2}{z^2 - r_p\left(e^{j\theta_p} + e^{-j\theta_p}\right)z + r_p^2} = \frac{z^2}{z^2 - r_p(2\cos\theta_p)z + r_p^2} = \frac{1}{1 - 2r_p\cos\theta_p z^{-1} + r_p^2 z^{-2}}$ **---(1)**
- ✓ $H_p(z) = \frac{1}{1 - b_1 z^{-1} + b_2 z^{-2}}$ $----- (2)$

Equating the denominator coefficients of (1) and (2) above, we obtain $\boldsymbol{b_1 = 2r_p \cos\theta_p}$ **and** $\boldsymbol{b_2 = r_p^2}$.

- ✓ Similarly, $a_1 = 2r_z \cos\theta_z$ and $a_2 = r_z^2$ for $H_z(z) = 1 - a_1 z^{-1} + a_2 z^{-2}$

$$H_p(z) = \frac{1}{1 - b_1 z^{-1} + b_2 z^{-2}}; \qquad H_z(z) = 1 - a_1 z^{-1} + a_2 z^{-2}$$

Resonator (Second-Order IIR filter)

Inverse Resonator (Second-Order FIR filter)

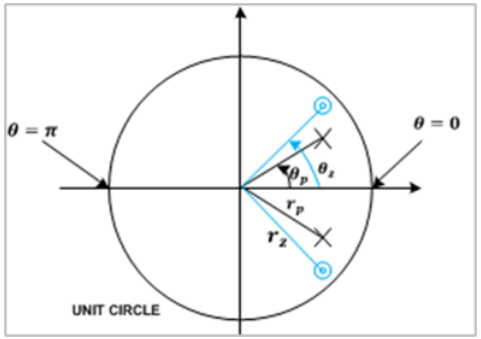
# Digital Filters: Normalisation and Design Example

- ✓ Normalise both filters to obtain unity gain at DC.
- ✓ $H_p(z) = \frac{1-b_1+b_2}{1-b_1z^{-1}+b_2z^{-2}}$ ; $H_z(z) = \frac{1-a_1z^{-1}+a_2z^{-2}}{1-a_1+a_2}$
- ✓ $r_p$ and $r_z$ can be calculated approximately as follows:

$$r_p \approx 1 - \left(\frac{BW_p}{f_s}\right)\pi \; ; \quad r_z \approx 1 - \left(\frac{BW_z}{f_s}\right)\pi$$

- ✓ Quality factors (Q – factors) can be calculated as follows:

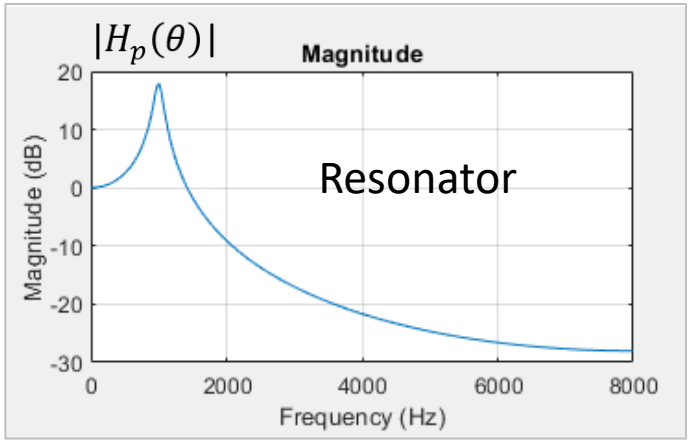
$$Q_p = \frac{f_p}{BW_p} \; ; \quad Q_z = \frac{f_z}{BW_z}$$

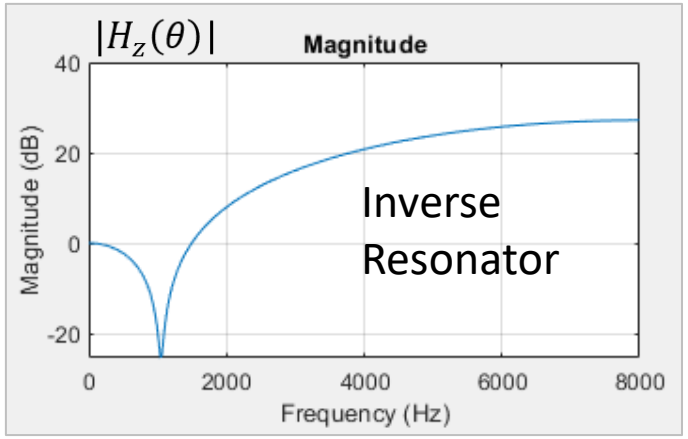



## Design Example

If $f_p$ = 1.0 kHz; $f_z$ = 1.05 kHz; $Q_p$ = 7.8 and $Q_z$ = 18.5, then

$$H_p(z) = \frac{0.1490}{1-1.8012z^{-1}+0.9503z^{-2}} \; ; \quad H_z(z) = \frac{1-1.8120z^{-1}+0.9778z^{-2}}{0.1659}$$



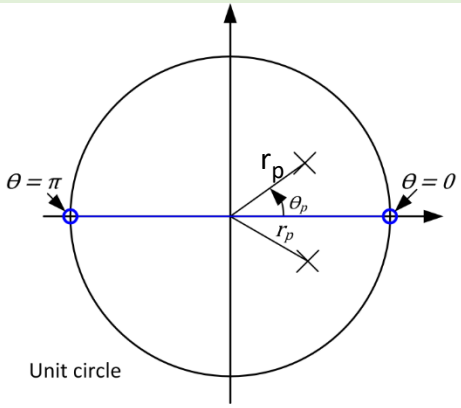
# Bandpass Filter Design Using Pole-zero placement

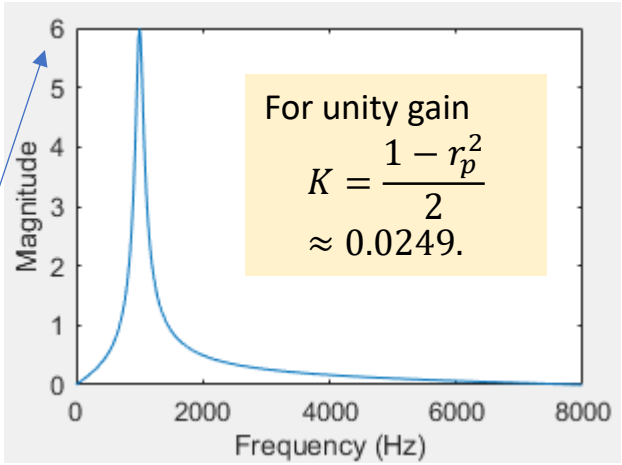
- ✓ Calculate the digital filter coefficients $b_1$ & $b_2$ and zeros using pole-zero placement (see the pole-zero diagram on the right):
  - Pole (angle) frequency = $\theta_p$; radius = $r_p$; $\theta_p = \frac{2\pi f_p}{f_s}$; $f_s = 16\ kHz$
  - $H_p(z) = \frac{K\,(z^2-1)}{\left(z-r_p e^{j\theta p}\right)\left(z-r_p e^{-j\theta p}\right)} = \frac{K\,(1-z^{-2})}{1-2r_p\cos\theta_p z^{-1}+r_p^2 z^{-2}}$ ----- (1)
  - $H_p(z) = \frac{K\,(1-z^{-2})}{1-b_1 z^{-1}+b_2 z^{-2}}$ ----- (2)
  - Equating the denominator of (1) and (2), we obtain

    $b_1 = 2r_p\cos\theta_p$ and $b_2 = r_p^2$ ----- (3)
  - $r_p$ and $BW_p$ (bandwidth of $H_p$(z)) can be calculated as follows:

    $r_p \approx 1 - \left(\frac{BW_p}{f_s}\right)\pi$ ; $BW_p = \frac{f_p}{Q_p}$; $Q_p$ - Q factor
- ✓ Calculate the denominator coefficients $b_1$ & $b_2$ using equation 3 above.
- ✓ If $f_p = 1\ kHz$ and $Q_p$ = 7.8 then $H_p(z) = \frac{0.1490\,(1-z^{-2})}{1-1.8012z^{-1}+0.9503z^{-2}}$
- ✓ K=0.1490 gives a peak magnitude of approximately **6**, consistent with the graph.

Complex-Conjugate Pole Pair and Zeros at z= $\pm 1$




For unity gain

$K = \frac{1-r_p^2}{2} \approx 0.0249.$

Magnitude response of a bandpass filter with centre frequency approximately 1000 Hz

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Term 3, 2026

# Digital Resonator Fundamentals: Stability of a Second-Order Filter

## Stability

- Consider a two-pole resonant filter given by:

$$H(z) = \frac{z^2}{z^2+b_1 z+b_2} = \frac{1}{1+b_1 z^{-1}+b_2 z^{-2}}$$ where $b_1$ & $b_2$ are the coefficients.

- This system has two zeros at the origin and poles at

$$p_1, p_2 = -\frac{b_1}{2} \pm \frac{\sqrt{{b_1}^2 - 4b_2}}{2}$$

- The filter is stable if the poles lie inside the unit circle i.e.

$$|p_1| < 1 \;\&\; |p_2| < 1$$

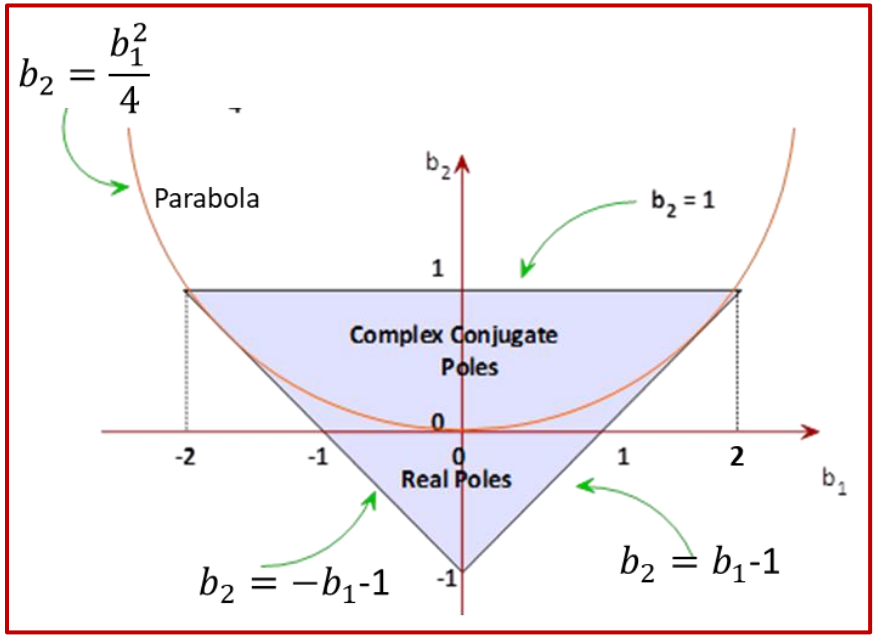
- For stability $-1 < b_2 < 1$ and $-1 - b_2 < b_1 < 1 + b_2$
- If $b_2 = 1$ then the system is an oscillator (**Marginally stable**)
- Assume that the poles are complex:

$${b_1}^2 - 4b_2 < 0 \;\Rightarrow\; {b_1}^2 < 4b_2 \quad \text{and} \quad b_1 < \pm 2\sqrt{b_2} \qquad b_2 > 0$$

$$b_1 = -2r\cos\theta_0 \quad \text{and} \quad b_2 = r^2$$

- If ${b_1}^2 - 4b_2 \geq 0$ then we get real roots. ${b_1}^2 = 4b_2$ *gives repeated real roots*
- The stability conditions define a region in the coefficient plane ($b_1$, $b_2$) which is in the form of a triangle (see diagram on the right)

A causal filter is stable if and only if the point (b1, b2) lies strictly inside the stability triangle.



# Digital Resonator Fundamentals: Stability of a Second-Order Filter

## Stability Triangle

- If the two poles ($p_1$ *and* $p_2$) are realeach must lie strictly between -1 and 1 for the causal system to be stable.

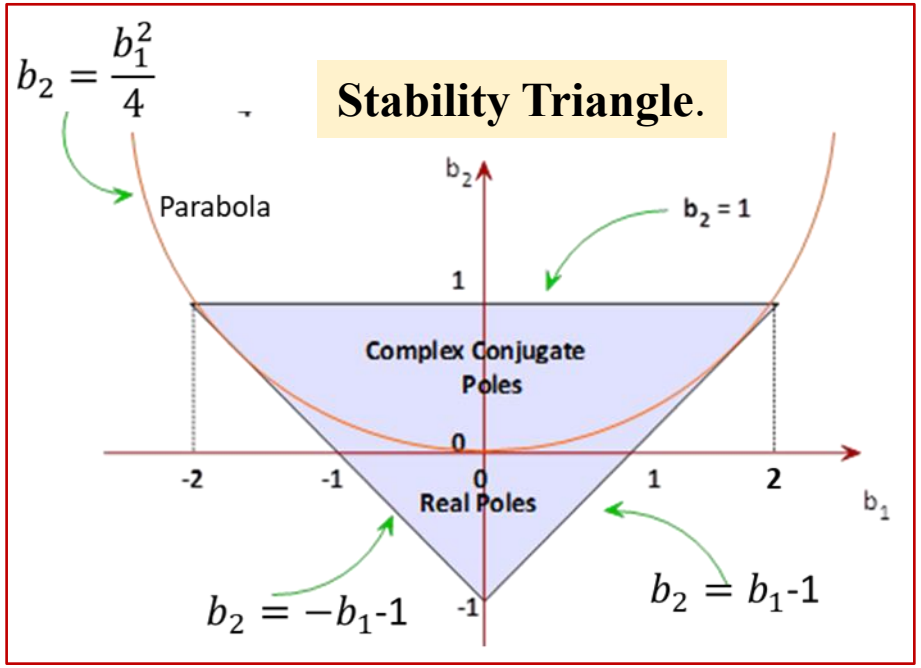
$$-1 < \frac{-b_1 \pm \sqrt{b_1^2 - 4b_2}}{2} < 1 \quad \Rightarrow \quad -2 + b_1 < \pm\sqrt{b_1^2 - 4b_2} < 2 + b_1$$

$$\therefore -2 + b_1 < -\sqrt{b_1^2 - 4b_2} \quad \text{and} \quad \sqrt{b_1^2 - 4b_2} < 2 + b_1$$

$$(-2 + b_1)^2 > b_1^2 - 4b_2 \quad \text{and} \quad b_1^2 - 4b_2 < (2 + b_1)^2$$

$$b_1 - b_2 - 1 < 0 \quad \text{and} \quad b_1 + b_2 + 1 > 0$$

- The region below the parabola ($b_1^2 > 4b_2$) corresponds to real and distinct poles.
- The points on the parabola ($b_1^2 = 4b_2$) result in real and equal (double) poles.
- The points above the parabola correspond to complex-conjugate poles.




For a stable 2nd order system, the coefficients must satisfy the following inequality:

$$-1 < b_2 < 1,$$
$$b_1 - b_2 - 1 < 0 \quad \& \quad b_1 + b_2 + 1 > 0$$

# ELEC3104: Digital Signal Processing

## Topic 4: Digital Resonator Fundamentals

- Second-Order Digital Resonator (Ref pp. 343-348)
- Stability of a Second - Order Filter (pp. 205-209)
- Digital Oscillators (Ref pp. 355-357)

Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

# Digital Resonator Fundamentals : Digital Oscillators

## Digital Oscillators

- A digital sinusoidal oscillator can be obtained from a second-order resonator by placing its complex-conjugate poles on the unit circle. $\boldsymbol{p_{1,2} = e^{\pm j\theta}}$ . The resulting oscillations have constant amplitude.
- The impulse response, h[n], of the second-order filter with complex conjugate **poles on the unit circle** is a sinusoid and the system is called a digital sinusoid / cosine oscillator.

$$h[n] = A\ r^n \cos[n\theta] = A\ cos[n\theta]\ u[n]$$ [poles on the unit circle, r=1]

- From the z-transforms table, the transfer function of a causal oscillator that generates a pure cosine wave $h[n] = A\ cos[n\theta]$ is given by ,

$$H(z) = \frac{Y(z)}{X(z)} = A\ \frac{1-(cos\ \theta)\ z^{-1}}{1-(2\ cos\ \theta)\ z^{-1}+z^{-2}}$$ [You may assume A =1]

- Taking inverse z-transform on both sides, we obtain

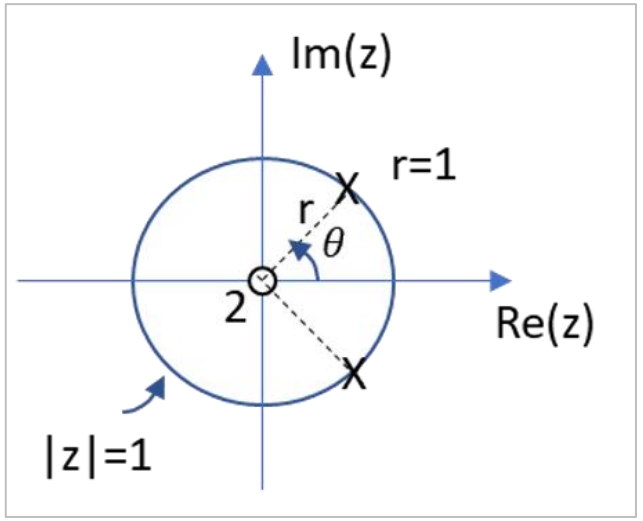
$$y[n] - 2\ cos\ \theta\ \ y[n-1] + y[n-2] = x[n] - cos\ \theta\ \ x[n-1]$$

- For an oscillator, the input $x[n] = 0\ and\ x[n-1]\ = 0$; So, the equation of the digital oscillator becomes:

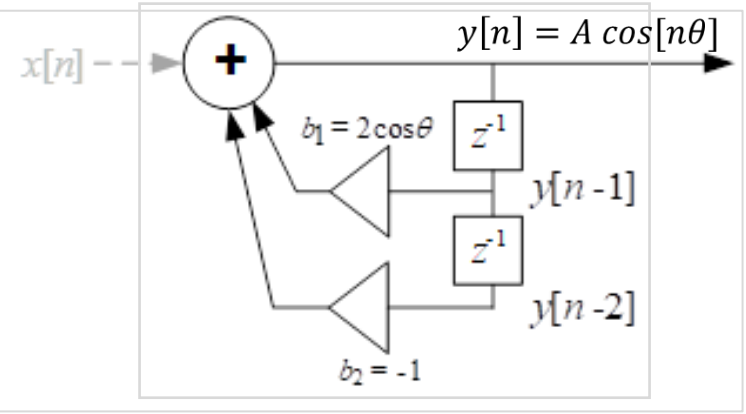
$$\boldsymbol{y[n] = 2\ cos\ \theta\ y[n-1] - y[n-2]}$$

and its structure is shown on the right:

Pole-zero plot




Digital Oscillator



# Digital Resonator Fundamentals : Digital Oscillators

## Initial conditions

**Case 1:** Assume the input x[n] = 0:

- To obtain $y[n] = A\cos(n\theta)$ , use the following initial conditions

$$y[0] = A\cos(0.\theta) = A \quad [for\ n=0]; \quad y[-1] = A\cos(-1.\theta) = A\cos\theta \quad [for\ n=-1]$$

- The frequency of the oscillator can be tuned by changing the coefficient $b_1$ ($b_2$ is a constant). The frequency $\theta$ of the oscillator is,

$$cos\,\theta = \frac{-b_1}{2\sqrt{b_2}} = -\frac{b_1}{2} \quad \text{[For an oscillator } b_2\text{=1]}$$

**Case 2:** Assume y[0]= 0 and y[-1]=0:

- In order to begin the sinusoidal oscillation, apply an impulse $\delta[n]$ at the input (see Figure A) when n= 0; Thereafter, the oscillation is self-sustaining because the system has no damping (i.e. r =1)

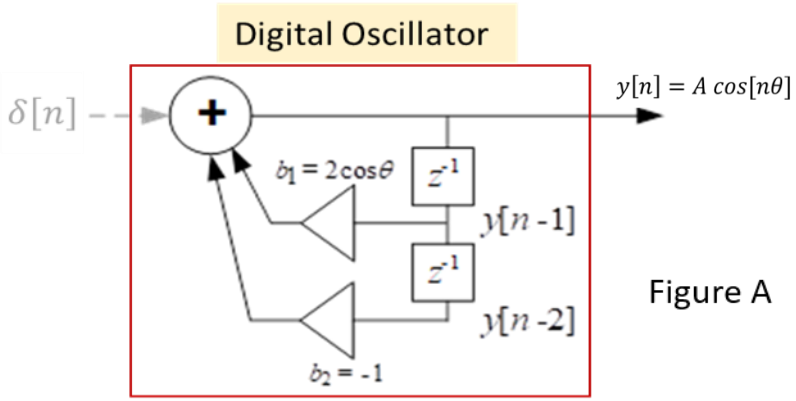



Figure A

## Digital Oscillator Applications

- The touch tone phone (DTMF phone: Dual tone multi frequency) makes use of digital oscillators to generate audible tones by pressing buttons on a keypad as shown in Figure B.
- Pressing a button produces a two-tone signal containing, a high and a low frequency tone
- Each button is associated with a unique pair of low-frequency and high-frequency tones. The tones can be generated by using a parallel combination of two programmable digital oscillators

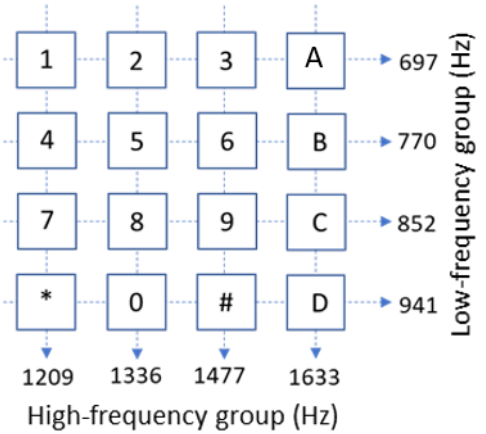
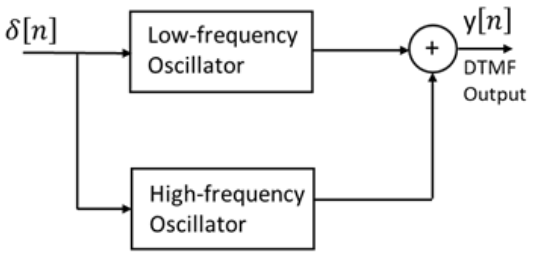



Figure B



# Digital Resonator Fundamentals: Digital Oscillators

Example 3:

- A digital sinusoidal oscillator is shown in Figure C :

(a) Assuming $\theta_0$ is the resonant frequency of the digital oscillator, find the values of $b_1$ and $b_2$ for sustaining the oscillation.

**Answer:**

$$H(z) = \frac{1}{1+b_1 z^{-1}+b_2 z^{-2}} \quad \text{----(1)}$$

$$H(z) = \frac{z^2}{(z - re^{j\theta_0})(z - re^{-j\theta_0})} = \frac{z^2}{z^2 - 2r\cos\theta_0\, z + r^2} \quad \text{---(2)}$$

By equating (1) and (2), we obtain: $b_1 = -2\, r\cos\theta_0;\ \ b_2 = r^2$

However, for oscillation (marginally stable system) , $b_2 = 1$ , $\therefore\ b_1 = -2\cos\theta_0$

(b)

(i) Write the difference equation corresponding to Figure C. You may assume that $x[n] = (A\sin\theta_0)\delta[n]$, and $y[-1] = y[-2] = 0$.

$$y[n] = -b_1 y[n-1] - y[n-2] + x[n]$$

$$y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\sin\theta_0\, \delta[n]$$

(ii) Show, by analysing the difference equation, that applying an impulse at $n$ = 0 initiates the sinusoidal oscillation and prove that the oscillation is self-sustaining thereafter.

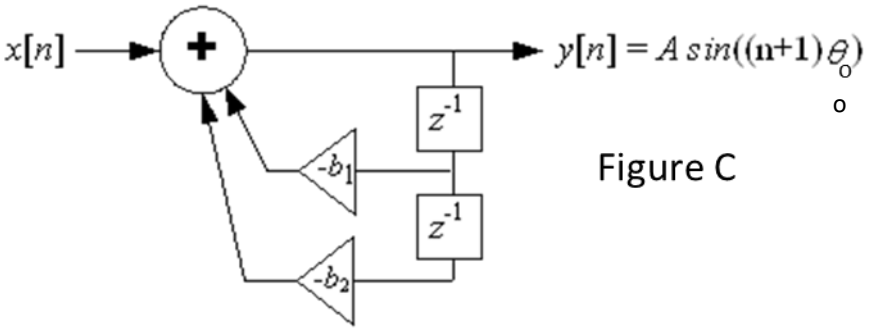



Figure C

# Digital Resonator Fundamentals : Digital Oscillators

Example 3:

$$y[n] = 2\cos\theta_0\, y[n-1] - y[n-2] + A\sin\theta_0\, \delta[n]$$

***Let n = 0***

$y[0] = 2\cos\theta_0\, y[-1] - y[-2] + A\sin\theta_0\, \delta[0]$ {$y[-1] = y[-2] = 0$; $\delta[0]=1$}

$\boldsymbol{y[0] = A\sin\theta_0}$

***Let n = 1***

$y[1] = 2\cos\theta_0\, y[0] - y[-1] + A\sin\theta_0\, \delta[1]$ {$y[-1] = 0$; $\delta[1]=0$}

$\boldsymbol{y[1] = 2\cos\theta_0\, A\sin\theta_0 = A\sin 2\theta_0}$

***Let n = 2***

$y[2] = 2\cos\theta_0\, y[1] - y[0] + A\sin\theta_0\, \delta[2]$ {$\delta[2] = 0$}

$= 2\cos\theta_0\, A\sin 2\theta_0 - A\sin\theta_0 = 2A\cos\theta_0\,[2\sin\theta_0\cos\theta_0] - A\sin\theta_0$

$= A\sin\theta_0\,[4\cos^2\theta_0 - 1] = A[3\sin\theta_0 - 4\sin^3\theta_0]$

$\boldsymbol{y[2] = A\sin 3\theta_0}$ and so forth.

$\therefore \boldsymbol{y[n] = A\sin(n+1)\theta_0}$ {The oscillation is self sustaining}

(c) By setting the input to zero and under certain initial conditions, sinusoidal oscillation can be obtained using the structure shown in Figure C. Find these initial conditions.

$y[n] = 2\cos\theta\, y[n-1] - y[n-2] + x[n]$ {$x[n]$ = 0 for an oscillator}

$y[n] = 2\cos\theta\, y[n-1] - y[n-2]$ $\boldsymbol{and}$ $\boldsymbol{y[n] = A\sin(n+1)\theta_0}$ $\boldsymbol{n \geq 0}$

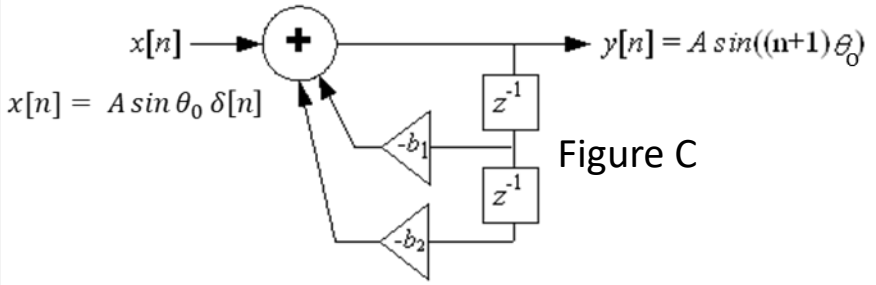



Figure C

Substitute n =-1 in $y[n] = A\sin(n+1)\theta_0$;
$\therefore$ y[-1] =0 - First initial condition.

Substitute n = -2 in $y[n] = A\sin(n+1)\theta_0$,
$\therefore$ y[-2] =$-A\sin\theta_0$ – Second initial condition.

# Digital Resonator Fundamentals : Digital Oscillators

## Sine and Cosine Oscillators

- Sinusoidal oscillators are normally used for generating carrier signals in modulation schemes .
- Some modulation schemes require both the sine and cosine signals. These signals can be generated simultaneously from the so-called coupled-form oscillator.
- Consider the trigonometric equation for $\cos(n\theta + \theta)$ :

$$\cos(n\theta + \theta) = \cos(n\theta)\cos(\theta) - \sin(n\theta)\sin(\theta)$$

Let $\boldsymbol{y[n]} = \boldsymbol{cos}[\boldsymbol{n}\theta]$ ; $y[0] = 0$.

$$\therefore y[n+1] = \cos(\theta)\, y[n] - \sin(\theta)\, x[n]$$

Replace *n* by *n*-1:

$$y[n] = \cos(\theta)\, y[n-1] - \sin(\theta)\, x[n-1] \quad \text{------- (1)}$$

- Similarly consider trigonometric equation for $\sin(n\theta + \theta)$

$$\sin(n\theta + \theta) = \sin(n\theta)\cos(\theta) + \cos(n\theta)\sin(\theta)$$

Let $\boldsymbol{x[n]} = \boldsymbol{sin}[\boldsymbol{n}\theta]$; $x[0] = 0$.

$$\therefore x[n+1] = \sin(\theta)\, y[n] + x[n]\cos(\theta)$$

Replace *n* by *n*-1:

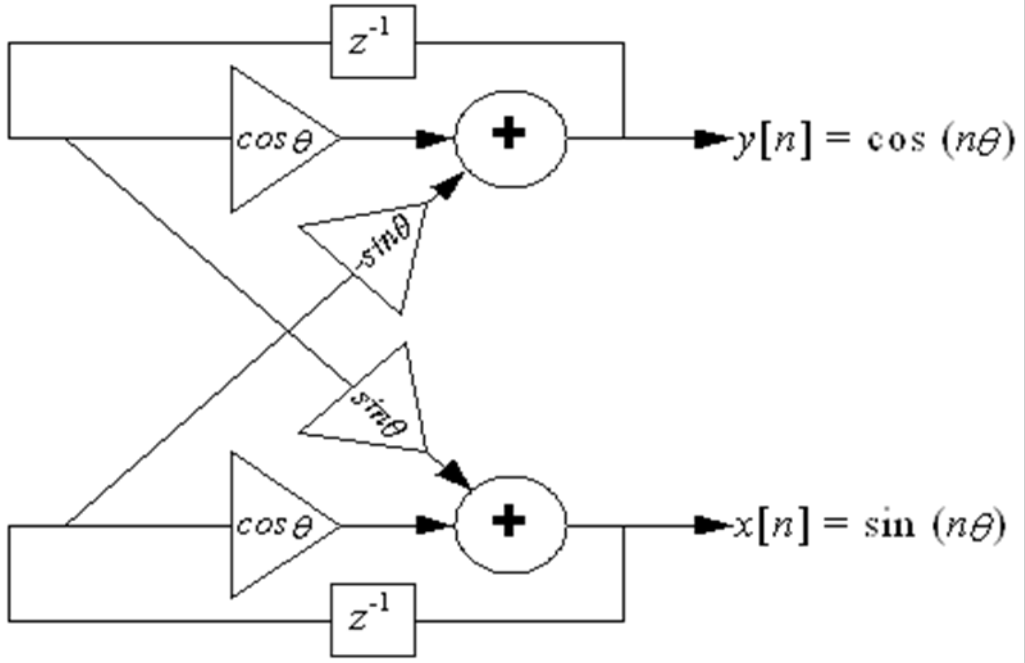
$$x[n] = \sin(\theta)\, y[n-1] + x[n-1]\cos(\theta) \quad \text{-------- (2)}$$

- Thus, we obtain the two coupled difference equations, where n ≥ 1.

$$y[n] = \cos(\theta)\, y[n-1] - \sin(\theta)\, x[n-1] \quad \text{--- (1)}$$

$$x[n] = \sin(\theta)\, y[n-1] + x[n-1]\cos(\theta) \quad \text{--- (2)}$$

- Using equations (1) and (2), the structure for the realisation of the coupled-form oscillator is shown below:



# Digital Resonator Fundamentals: Digital Oscillators

Example 4:

- An oscillator is given by the following coupled difference equations expressed in matrix form.

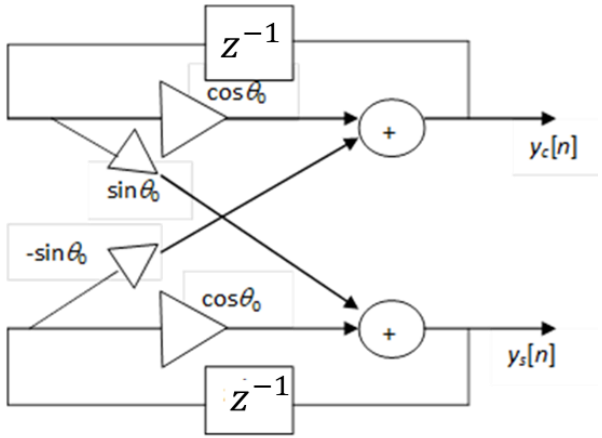
$$\begin{bmatrix} y_c[n] \\ y_s[n] \end{bmatrix} = \begin{bmatrix} cos\,\theta_0 & -sin\,\theta_0 \\ sin\,\theta_0 & cos\,\theta_0 \end{bmatrix} \begin{bmatrix} y_c[n-1] \\ y_s[n-1] \end{bmatrix}$$

(a) Draw the structure for the realisation of this oscillator, where $\theta_0$ is the oscillation angular frequency.

(b) If the initial conditions are $y_c[-1] = Acos\theta_0$ and $y_s[-1] = -Asin\theta_0$, obtain the outputs $y_c[n]$ and $y_s[n]$ using the above difference equations.

**Answer:**

$$y_c[n] = cos\,\theta_0\, y_c[n-1] - sin\,\theta_0\, y_s[n-1]$$
$$y_s[n] = sin\,\theta_0\, y_c[n-1] + cos\,\theta_0\, y_s[n-1]$$




(b)

$\boldsymbol{Let\ n = 0}$

$y_s[0] = \sin\theta_0\,(A\cos\theta_0) + \cos\theta_0\,(-Asin\theta_0) = 0$

$y_c[0] = \cos\theta_0\,(Acos\theta_0) - \sin\theta_0\,(-Asin\theta_0) = \boldsymbol{A}$

$\boldsymbol{Let\ n = 1}$

$y_c[1] = \cos\theta_0 . A - \sin\theta_0 .0 = \boldsymbol{Acos[1\ \theta_0]}$

$y_s[1] = A\sin\theta_0 + 0 = A\sin\theta_0$

**Let $\boldsymbol{n = 2}$**

$y_c[2] = \cos\theta_0\, yc[1] - \sin\theta_0\, ys[1]$

$= \cos\theta_0\, A\cos\theta_0 - \sin\theta_0\, A\sin\theta_0$

$= \boldsymbol{A}\cos[\mathbf{2}\ \theta_0]$

**Let $\boldsymbol{n = n}$**

$y_c[n] = \boldsymbol{A}\cos[\boldsymbol{n}\ \theta_0]$

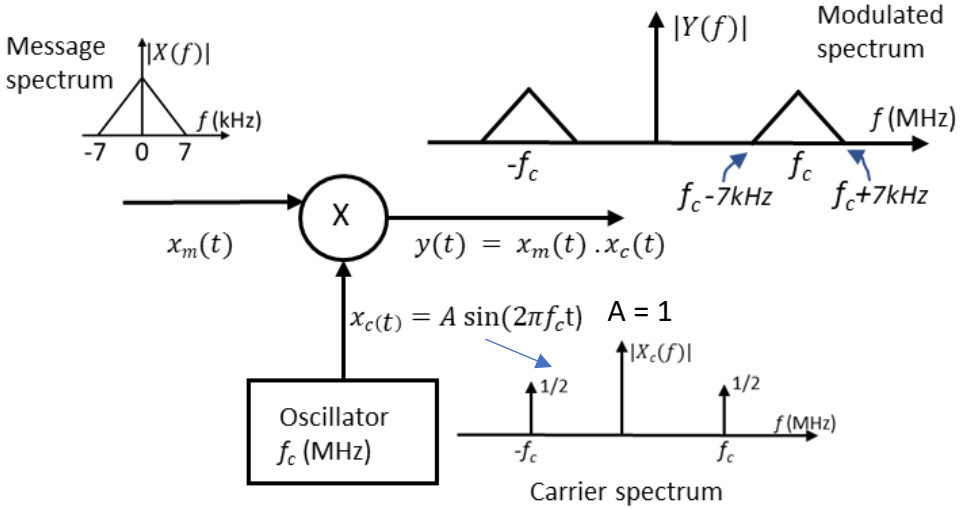
$Similarly \quad y_s[n] = A\sin[n\theta_0]$

# Digital Oscillator Applications: Amplitude Modulation

Note:

- Communication of information often requires transmission of message through space by converting electrical signals to electromagnetic waves using antennas.
- Efficient radiation requires antenna dimensions comparable to the wavelength of the signal .
- The wavelength $\lambda$ of a radiated signal equals $\frac{c}{f}$ , where c is the speed of light (at which electromagnetic waves travel) and $f$ is the signal frequency.
- Transmission of low frequency messages, such as speech and music, would require huge antenna sizes (in km - normally $\lambda/2$)
- A practical approach is to shift low-frequency messages to a higher frequency using *modulation* and to transmit the modulated signal instead.
- In turn, this requires a means of recovering the original message at the receiving end using a process called *demodulation*.
- Such modulation Scheme (known as amplitude modulation) is shown on the right, where the carrier frequency is generated by an oscillator.

Double-Sideband Suppressed-Carrier (DSB-SC) Modulation




- Amplitude modulation uses a message signal to vary the amplitude of a high-frequency carrier.
- In the above scheme (Double-sideband suppressed carrier), the carrier $x_c(t)$ is multiplied by the message $x_m(t)$ .
- The message spectrum is centered about $\pm f_c$ as shown above.
- At the receiver, the message $x_m(t)$ can be recovered using demodulation

# Topic 4 - Summary

## Summary

By the end of Topic 4, you should be able to:

- Derive the transfer function of a second-order resonator and analyse its stability using pole positions and the stability triangle.
- Determine the coefficient conditions required for filter stability.
- Distinguish between stable, marginally stable and unstable systems, and derive the difference equations for digital oscillators.
- Draw and correctly initialise a coupled-form oscillator that generates sine and cosine signals simultaneously.
- Design notch filters, bandpass filters and resonators using pole–zero placement.
- Design combinations of resonators and notch filters and analyse their magnitude responses.